

Advaith Appajodu

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. in Applied Mathematics and Computer Science

2028

- **Relevant Coursework:** Data Structures, Structure & Interpretation of Computer Programs, Multi-variable Calculus, Linear Algebra & Differential Equations, Discrete Mathematics, Intro to Quant Finance

EXPERIENCE

Berkeley AI Research (BAIR) Lab – Jiao Group

Oct. 2025 – Present

Student Researcher (Advisor: Prof. Jiantao Jiao)

Berkeley, CA

- Designed and deployed **scalable evaluation pipelines** for generative video models, benchmarking outputs on physical realism, temporal consistency, and constraint satisfaction.
- Built **model evaluation infrastructure** to quantify alignment with real-world physics, enabling systematic comparison across architectures and training regimes.

Salesforce AI Research (Mentorship)

Jan. 2024 – June 2024

Research Mentee

San Francisco, CA

- Investigated **LLM fine-tuning and adaptation techniques** (LoRA) to improve function-calling reliability in enterprise-scale systems.
- Developed a **high-throughput data processing pipeline** using Hugging Face to clean, normalize, and structure large-scale training datasets.
- Gained experience with **distributed training workflows**, dataset curation, and evaluation strategies for production LLM systems.

Cursor Explore Program

March 2026

- Selected for a competitive immersion program, shadowing **Engineering, ML, and GTM teams** across product development and deployment workflows.
- Observed real-world **developer workflows and agentic automation systems**, analyzing how LLM-powered tools improve software engineering productivity.
- Studied **ML training, evaluation, and feedback loops** in production environments, including iteration cycles and model performance tradeoffs.

PROJECTS

High-Performance Deep Learning Systems | *PyTorch, CUDA, CNNs, Transformers*

- Built and trained deep learning models from scratch, including CNNs (CIFAR-10) and Transformers for machine translation using multi-head attention and positional encodings.
- Achieved **82% test accuracy on CIFAR-10** via architecture design, regularization, and optimizer tuning (AdamW, BatchNorm, data augmentation).
- Trained an English–Telugu Transformer achieving **76 BLEU**, implementing teacher forcing, tokenization pipelines, and end-to-end training with logging (Weights & Biases).

trAId | *Python, FastMCP, Redis, Browserbase, React, TypeScript, Node.js*

- Won 2026 **CalHack's AI Hackathon** Sponsor Track
- Built an **autonomous resale agent** that cuts pricing and listing an item from **hours to seconds**, operating through the user's texts via Poke and using sent photos to identify, price, and list objects automatically.
- Engineered a **Python/FastMCP server** backed by Redis and Browserbase for headless browser automation, exposed to Poke as a **custom MCP integration** to drive end-to-end listing workflows across marketplaces.
- Architected a **real-time React + TypeScript dashboard** with a Node/Express layer streaming live agent events over **SSE**, and integrated MCP tooling for automated eBay listing creation.

TECHNICAL SKILLS

Languages: Python, C++, Java, Go, SQL, JavaScript/TypeScript

ML/AI: PyTorch, Transformers, LoRA, Hugging Face, MCP, Model Evaluation, Training Pipelines

Systems: Distributed Training, Data Pipelines, GPU Workloads, Benchmarking, Observability

Web: React, Next.js, REST APIs, Mapbox, SQLite

Tools: Docker, Git, Weights & Biases, Browserbase, Redis